

SQL (Structured Query Language)

Operations on Table :-

➤ SELECT Command :-

- SQL SELECT is used to fetch or retrieve data from a database.
- It can fetch all the data from a table or return specific results based on specified conditions.
- The data returned is stored in a result table.

1) SELECT AS

SELECT AS

- **SQL AS** is used to assign temporarily a new name to a table column.
- Allows the developer to label results more accurately without permanently renaming table columns.

Syntax

SELECT column_name **AS** column_name **FROM** table_name;

Eg. **select** username **as** 'name' **from** student;

2) SELECT IN

SELECT IN

- SQL IN is an operator used in a SQL query to help reduce the need to use multiple SQL "OR" conditions.
- It minimizes the use of SQL OR operator.

Syntax

SELECT * FROM table_name **WHERE** column_name **IN** (val1,val2,...);

Eg. **select * from** student **where** name **in**('abc','lmn');

3) SELECT SUM

SELECT SUM

- It is also known as SQL SUM() function.
- It is used in a SQL query to return summed value of an expression.

Syntax

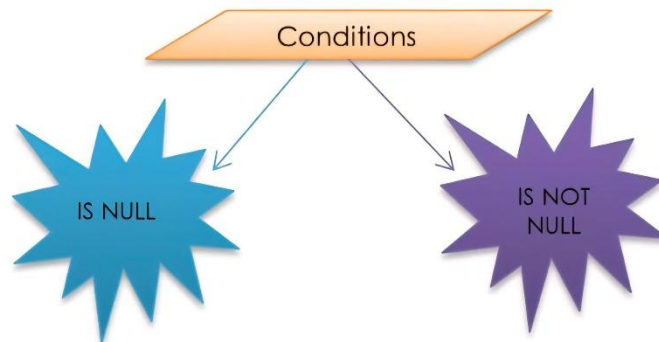
SELECT SUM (column_name) **AS** "expression_name" **FROM** table_name;

Eg. **select sum**(salary) **as** "total salary" **from** employee;

4) SELECT NULL

SELECT NULL

- Null values are used to represent missing unknown data.



I. IS NULL

IS NULL

- Select records with null values.

Syntax

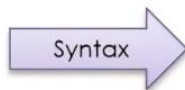
SELECT column_name **FROM** table_name **WHERE** column_name **IS NULL**;

Eg. **select** name **from** student **where** marks **is null**;

II. IS NOT NULL



- Select records with no null values.



SELECT column_name **FROM** table_name **WHERE**
column_name **IS NOT NULL**;

Eg. **select** name **from** student **where** marks **is not null**;

➤ CLAUSES :-

- SQL clauses are built-in functions that define specific conditions within an SQL statement to retrieve, update, or manipulate data from a database.
- Here are some of the most common types of clauses in SQL :



1) WHERE CLAUSE



- The SQL **WHERE** clause is used to specify a condition while fetching the data from a single table or by joining with multiple tables.
- You should use the WHERE clause to filter the records and fetching only the necessary records.



SELECT * FROM table_name **WHERE** condition;

Eg. **select * from** student **where** id=1;

2) AND CLAUSE

AND CLAUSE

- The SQL **AND** condition is used in SQL query to create two or more conditions to be met.

Syntax

SELECT * FROM table_name **WHERE** condition1 **AND** condition2;

Eg. **select * from** student **where** id=1 **AND** name='abc';

3) OR CLAUSE

AND CLAUSE

- The SQL **AND** condition is used in SQL query to create two or more conditions to be met.

Syntax

SELECT * FROM table_name **WHERE** condition1 **AND** condition2;

Eg. **select * from** student **where** id=1 **AND** name='abc';

4) ORDER BY CLAUSE

- The SQL ORDER BY clause is used for sorting data in ascending and descending order based on one or more columns.
- Databases sort query results in ascending order by default.

ASC

DESC

RANDOM

LIMIT

I. ASC

ASC

- It is used to sort data in ascending order.
- SQL ORDER BY query takes ascending order by default.

Syntax

SELECT * FROM table_name ORDER BY column_name ASC;

Eg. **select * from student ORDER BY id ASC;**

II. DESC

DESC

- It is used to sort data in descending order.

Syntax

SELECT * FROM table_name ORDER BY column_name DESC;

Eg. **select * from student ORDER BY id DESC;**

III. RANDOM & LIMIT

RANDOM

LIMIT

- If you want to fetch random rows from any of the databases you have to use **RANDOM()**
- **SQL LIMIT** - We can retrieve limited rows from the database.

Syntax

SELECT * FROM table_name ORDER BY RAND() LIMIT 1;

Eg. **select * from student order by RAND() LIMIT 1;**